ANITHA GUDIMETLA
SENIOR DATA ENGINEER

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PROFESSIONAL SUMMARY:

- Having around **8+ years** of experience in the IT industry as a **Data Engineer** with proven expertise in designing and implementing data solutions involving cloud computing platforms like AWS and Azure.
- Extensively worked on **AWS** Cloud services like **EC2, EMR, ECS, Auto Scaling, S3, CloudFront, Elastic Beanstalk, Lambda, Elastic Cloud Formation, Redshift, DynamoDB, SNS, SQS, SES, Lambda, Cognito IAM.**
- Experience on various **Azure** Services like **Azure SQL Database, Azure Data Factory, Azure Blob Storage, Azure Data Lake Storage, Cosmos DB, Azure Virtual Network, Azure VPN Gateway, Compute, Azure Active Directory, Azure Functions, API Management, Azure Logic Apps, Azure Virtual Machine Scale Sets.**
- Experience in migrating on-premises to **Azure** and building **Azure Disaster Recovery** Environment and **Azure backups** from scratch using **PowerShell script**.
- Proficient in designing and implementing **ETL** processes using industry-leading tools like **Informatica**, **Talend**, and **Apache NiFi,** ensuring efficient data extraction, transformation, and loading into **data warehouses**.
- Expertise in the full life cycle of ETL (**Extraction, Transformation, and Loading**) using **Informatica** power-center and **Apache Airflow**
- Skilled in optimizing data integration workflows using **Matillion**.
- Expert in **AWS Glue** for **ETL (Extract, Transform, Load)** operations, optimizing data workflows, and integrating with various **AWS services**.
- Skilled in designing and implementing data integration and transformation workflows using **Azure Data Factory (ADF).**
- Skilled in building **machine learning models** in **Python** using libraries like **Scikit-Learn and TensorFlow.**
- Experienced in optimizing **PySpark jobs** for performance improvements and scalability in distributed computing environments.
- Expert in writing complex **SQL queries** and optimizing performance using **SparkSQL** for large-scale data processing.
- Used **Spark SQL for Scala & amp**, **Python** interface that automatically converts **RDD** case classes to schema RDD.
- Excellent knowledge of **Hadoop** components such as **HDFS, Job Tracker, Task Tracker, Name Node, Data Node, YARN, and MapReduce** programming paradigm.
- Developed and documented **Control-M job flows**, dependencies, and scheduling policies.
- Expert in designing and optimizing **MS SQL Server databases** for high-performance data processing.
- Experience in using and tuning relational databases like **Microsoft SQL Server, Oracle, MySQL** and columnar databases like **Amazon Redshift, Microsoft SQL Data Warehouse**
- Hands-on experience with **Snowflake's** unique architecture including virtual warehouses.
- Experienced in leveraging **machine learning algorithms** and **statistical models** to extract actionable insights from complex datasets.
- Capable of optimizing **Power BI** performance by tuning queries, data models, and report design for enhanced user experience.
- Conducted **Data blending, Data preparation** using **Alteryx** and **SQL** for **Tableau** consumption and publishing data sources to **Tableau server.**
- Actively contribute to open-source projects on **GitHub**.
- Familiar with **GitLab's** issue tracking, project management, and collaboration tools.
- Actively utilize **Azure DevOps** for continuous monitoring and improvement of data workflows, ensuring scalability and reliability of data solutions.
- Capable of integrating **Jira** with other tools like **Confluence** and **Bitbucket** to enhance communication and documentation within cross-functional teams.
- Seasoned Leader with a proven track record of leading cross-functional teams in the successful delivery of large-scale data projects, utilizing **agile** methodologies to ensure timely and within-budget completion while maintaining high standards of quality.
- **Proactive Problem Solver** with a track record of identifying data pipeline inefficiencies and implementing efficient solutions to improve data reliability and accessibility.

- **Effective Communicator** skilled in translating complex technical concepts into actionable insights for diverse stakeholders, fostering a collaborative environment for achieving project goals.

TECHNICAL SKILLS:

Programming Languages	Python, Scala, PySpark, SQL, PowerShell, Java, R
Data Integration	AWS Glue, Informatica, Apache Airflow, Matillion
Big Data Technologies	Hadoop, Spark, Hive, HiveQL
Data Warehousing	AWS Redshift, Azure Synapse Analytics, Snowflake, Google BigQuery, Oracle Exadata
Database Management	MS SQL Server, MySQL, PostgreSQL, MariaDB, Oracle
BI & Visualization Tools	Power BI, Tableau, QlikView, Google Analytics, SAS
Version Control & CI/CD	Git, GitLab, GitHub Actions, Azure DevOps, CI/CD pipelines, Jupyter Notebook
Machine Learning	Scikit-Learn, TensorFlow, SAS, R
Other Tools & Technologies	Control-M, Jira, Confluence, Bitbucket, Excel (VLOOKUP, INDEX-MATCH, pivot tables), Alteryx

PROFESSIONAL EXPERIENCE:

Client: Connecticut Department of Transportation, Newington, CT. **Oct 2022 - Present**
Role: Senior Data Engineer
Responsibilities:

- Managed **Azure** data services including **Azure Data Factory, Azure Databricks, and Azure Synapse Analytics** for efficient **ETL** pipelines and data integration.
- Implemented **Azure SQL Database** for **data warehousing solutions**, optimizing performance and scalability.
- Utilized **Azure Blob Storage and Azure Data Lake Storage** for storing and managing big data sets securely.
- Designed and implemented **ETL** pipelines using **Azure Data Factory** to orchestrate data movement and transformation across various data sources.
- Utilized **Azure Synapse Analytics** to integrate data from multiple sources, enabling comprehensive and unified analytics.
- Implemented robust data validation and cleansing processes within **Azure Data Factory** to ensure high data quality and reliability.
- Configured and optimized **Azure Event Hubs** for real-time data ingestion, ensuring low-latency data processing and integration.
- Developed custom **Azure Functions** to handle complex data transformations and processing tasks within the **ETL** pipelines, improving efficiency and scalability.
- Implemented real-time data processing solutions using **Apache Spark and Spark Streaming**, leveraging **Scala** and **Python programming languages.**
- Implemented and optimized complex **SQL** queries, stored procedures, and functions in **MySQL** to enhance data retrieval performance and support extensive data processing workflows.
- Developed advanced **Python** scripts for automating data extraction, transformation, and loading (ETL) processes, leveraging libraries such as **Pandas**, **NumPy**, and **SQLAlchemy** to streamline data pipeline operations.
- Implemented robust data processing and analytics solutions using **PySpark**, including designing scalable **ETL** workflows, performing data aggregations.
- Import the data from different sources like **HDFS/HBase into Spark RDD** and perform computations using **PySpark** to generate the output response.
- Performed data aggregation and joined operations across multiple data sources using **SparkSQL**.
- Integrated external libraries and frameworks into **Scala Spark** applications for enhanced functionality.
- Extracted and transformed data from multiple sources into **Power BI** datasets for unified analytics.
- Implemented row-level security and data-level security in **Power BI** to ensure data integrity and confidentiality.
- Used **GitHub** Actions for automating build, test, and deployment pipelines.
- Utilized **Jira** reporting capabilities to generate actionable insights and metrics, improving project visibility and decision-making across data engineering initiatives.

Environment: Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Azure SQL Database, Azure Blob Storage, Azure Data Lake Storage, Azure Event Hubs, Azure Functions, Apache Spark, Spark Streaming, Scala, Python, MySQL, Pandas, NumPy, SQLAlchemy, PySpark, SparkSQL, HDFS, HBase, Power BI, GitHub, Jira.

Client: Bank of America, Charlotte, NC. **May 2019 – Sep 2022**

Role: Senior Data Engineer

Responsibilities:

- Managed and optimized **AWS infrastructure, including EC2 instances, S3 storage, and IAM** roles for scalable data processing workflows.
- Utilized **AWS Glue** for **ETL** processes, ensuring efficient data transformation and integration across heterogeneous data sources.
- Configured **AWS Redshift clusters** and optimized queries to support data analytics and reporting requirements.
- Implemented monitoring and logging solutions on **AWS CloudWatch** to ensure high availability and performance of data workflows and applications.
- Experienced in working with **AWS Redshift, Cloudera, and utilizing Spark, Pyspark**, and **Hadoop** for data processing and analysis.
- Implemented and implemented serverless data processing pipelines using **AWS Lambda, S3, and DynamoDB** to achieve scalable and cost-effective solutions.
- Led the migration of on-premises data warehouses to **Amazon Redshift**, reducing operational costs and improving query performance by 30%.
- Implemented data lake solutions on **AWS** using **S3**, **Glue**, and **Athena**, enabling ad-hoc querying and analysis of large datasets without extensive data movement.
- Automated data backup and recovery processes using **AWS Backup** and **S3** versioning to ensure data integrity and availability in case of failures.
- Conducted thorough testing and **debugging** of **ETL** processes to identify and resolve data integrity issues, ensuring accurate and reliable data delivery.
- Optimized performance of **Informatica** mappings and workflows by tuning **SQL queries**.
- Prepared scripts to automate the ingestion process using **Python and Scala** as needed through various sources such as **API, AWS S3, Teradata and snowflake.**
- Optimized **PySpark** scripts for performance improvements, leveraging **RDDs** and **DataFrames**.
- Developed **SparkSQL** scripts for data validation and integrity checks.
- Optimized **Scala Spark jobs** using **RDDs, DataFrames, and Spark SQL** for optimal performance.
- Monitored **Hadoop cluster** performance and capacity planning to scale infrastructure as needed.
- Developed and implemented **Control M** workflows to orchestrate **ETL** processes across multiple environments.
- Developed and maintained **data models** for relational databases including **Oracle** and **MS SQL Server**, optimizing performance and ensuring data integrity.
- Optimized **MySQL** queries and indexes for improved performance and response times.
- Evaluated and integrated third-party **machine learning libraries** and frameworks to enhance project capabilities.
- Developed interactive dashboards and reports in **Tableau**, enabling stakeholders to visualize and explore complex datasets, driving data-informed decision-making.
- Integrated **Tableau** with various data sources, including **AWS Redshift** and **MySQL**, to create seamless and real-time data visualizations for business intelligence.
- Collaborated with cross-functional teams to review and merge code changes using **GitHub** pull requests.
- Ensured compliance and security standards in **GitLab** repositories and **CI/CD pipelines.**
- Conducted **Jira** training sessions for team members to enhance proficiency in **Agile** project management practices and tool utilization.

Environment: AWS EC2, AWS S3, IAM roles, AWS Glue, AWS Redshift, AWS CloudWatch, Cloudera, Spark, PySpark, Hadoop, AWS Lambda, DynamoDB, AWS Backup, S3 versioning, Informatica, SQL, Python, Scala, API, Teradata, Snowflake, RDDs, DataFrames, Spark SQL, Hadoop cluster, Control M, Oracle, MS SQL Server, MySQL, third-party machine learning libraries, Tableau, GitHub, GitLab, CI/CD pipelines, Jira.

Client: TATA AIG General Insurance Company Limited, Mumbai, India. **Feb 2017 – Mar 2019 Role: Data Engineer Responsibilities:**

- Developed and maintained **ETL** processes leveraging **Azure Databricks and Apache Spark.**
- Ensured **data security** and compliance by configuring **Azure Key Vault** and **Role-Based Access Control.**
- Monitored and troubleshot data workflows using **Azure Monitor and Azure Log Analytics.**

- Implemented data validation and quality checks using **Azure Data Factory** to ensure accuracy and consistency in **ETL** processes.
- Utilized **Azure Synapse Analytics** to integrate and analyze large-scale datasets, providing actionable insights to business stakeholders.
- Configured **Azure Data Lake Storage** for scalable and secure storage of structured and unstructured data.
- Ensured compliance with **data governance** and security policies while designing and developing Informatica workflows for sensitive data handling.
- Designed and developed **Talend** jobs to ingest data from diverse sources into the **data lake.**
- Collaborated with data scientists to deploy **machine learning models** into production using **Python** frameworks like **scikit-learn and TensorFlow.**
- Tuned **Spark** jobs for performance improvements, leveraging **PySpark's** caching and partitioning strategies.
- Designed and optimized **SQL queries** in **SparkSQL** for querying large-scale datasets distributed across clusters. Implemented fault-tolerant data pipelines in **Scala Spark** to handle data ingestion from various sources.
- Implemented **Hadoop MapReduce jobs** to process and analyze large-scale datasets stored in **HDFS**.
- Developed and maintained **Control-M** workflows for **data ingestion, transformation, and loading processes.**
- Managed **MySQL replication** and performed database performance tuning to enhance query speed.
- Leveraging **Snowflake's** cloud data warehouse for multi-terabyte datasets, improving query performance and reducing storage costs by **30%.**
- Optimized **Snowflake database** performance using clustering keys and materialized views.
- Developed and maintained **Tableau dashboards** to visualize key business metrics and trends for stakeholders.
- Automated deployment workflows using **Git hooks** and **CI/CD pipelines** for efficient data pipeline updates.
- Collaborated with cross-functional teams to maintain **Jira** boards, ensuring clear visibility and transparency throughout project lifecycles.

Environment: Azure Databricks, Azure Key Vault, Azure Monitor, Azure Log Analytics, Azure Data Factory, Azure Synapse Analytics, Azure Data Lake Storage, Apache Spark, Role-Based Access Control (RBAC), Informatica, Talend, Python, scikit-learn, TensorFlow, PySpark, SparkSQL, Scala Spark, Hadoop MapReduce, HDFS, Control-M, MySQL, Snowflake, Tableau, Git, CI/CD, Jira

Client: Medika bazaar, Mumbai, India. **Jul 2016 – Jan 2017**
Role: Data Analyst
Responsibilities:

- Leveraged **AWS Glue** for serverless data integration and **ETL tasks** within the **AWS ecosystem.**
- Monitored and optimized **data pipelines** using **AWS CloudWatch** for performance and reliability.
- Implemented data warehousing solutions with **AWS Redshift** and **Athena** for scalable analytics.
- Developed and maintained **ETL** processes using Informatica to integrate data from various sources.
- Utilized **SQL** for complex **data queries, transformations, and report generation.**
- Created interactive dashboards and visualizations in **QlikView** for business insights.
- Conducted **data analysis and statistical modeling** with **Python, R, and SAS**.
- Designed and implemented **Star Schema data models** for optimized querying and reporting.
- Managed **big data processing** using **Hadoop**, **PySpark, and Apache Spark** for large datasets.
- Automated data workflows with **Airflow** and **Cron jobs** to ensure timely data processing.
- Developed data processing scripts using **Jupyter Notebook** with **Python** (**Pandas**, **NumPy**) for exploratory data analysis and testing, optimizing workflows for data preprocessing and visualization.
- Designed, optimized, and managed relational database schemas in **MariaDB**, **MySQL**, **PostgreSQL**, and **Oracle** Exadata, ensuring efficient data storage and retrieval for large-scale structured data.
- Leveraged advanced Excel functions such as **VLOOKUP, HLOOKUP, INDEX-MATCH,** and pivot tables to manipulate and analyze large datasets, streamlining financial reporting and decision-making processes.
- Utilized **Hive** and **HiveQL** to efficiently query and manage multi-terabyte datasets in distributed **Hadoop** environments, optimizing data retrieval processes for big data analytics.
- Employed **SAS** for advanced statistical analysis and predictive modeling.

Environment: AWS Glue, AWS CloudWatch, AWS Redshift, AWS Athena, Informatica, SQL, QlikView, Python, R, SAS, Star Schema, Hadoop, PySpark, Apache Spark, Airflow, Cron jobs, Jupyter Notebook, Pandas, NumPy, MariaDB, MySQL, PostgreSQL, Oracle Exadata, Excel, VLOOKUP, HLOOKUP, INDEX-MATCH, pivot tables, Hive, HiveQL

EDUCATION: Jawaharlal Nehru Technology University, Hyderabad, TS, India BTech in Computer Science and Engineering, June 2012 - May 2016